# Task 2: Data Visualization and Storytelling

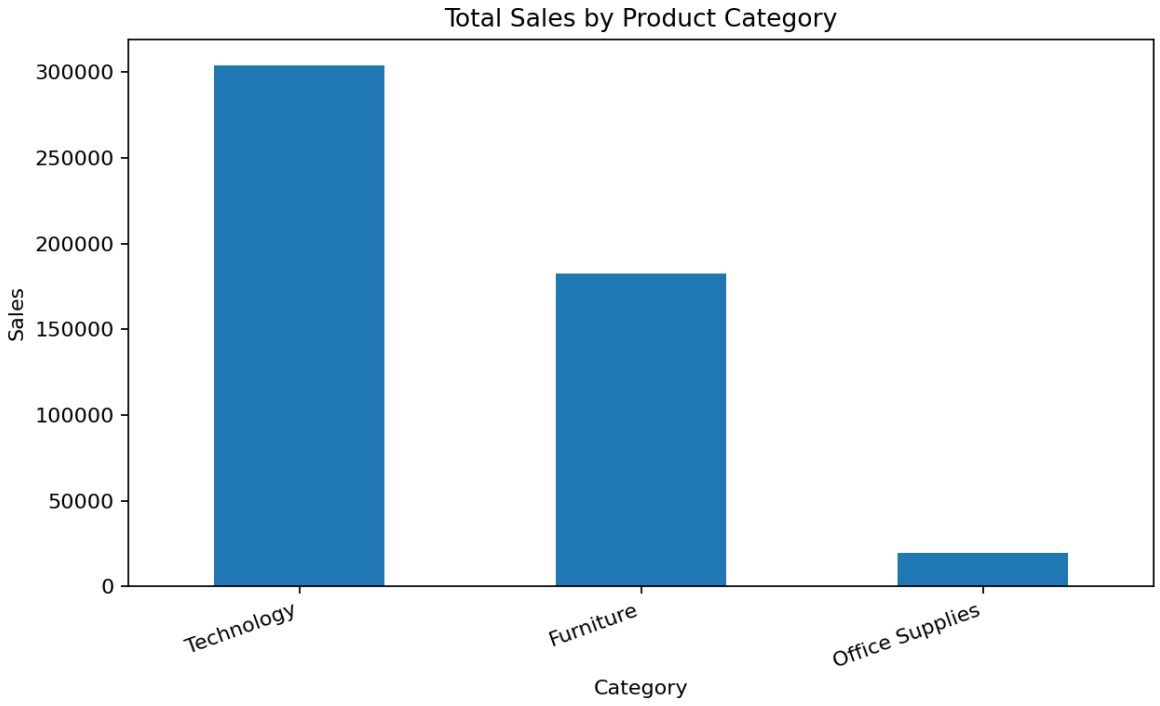
## Data Analyst Internship – Sample Sales Analysis

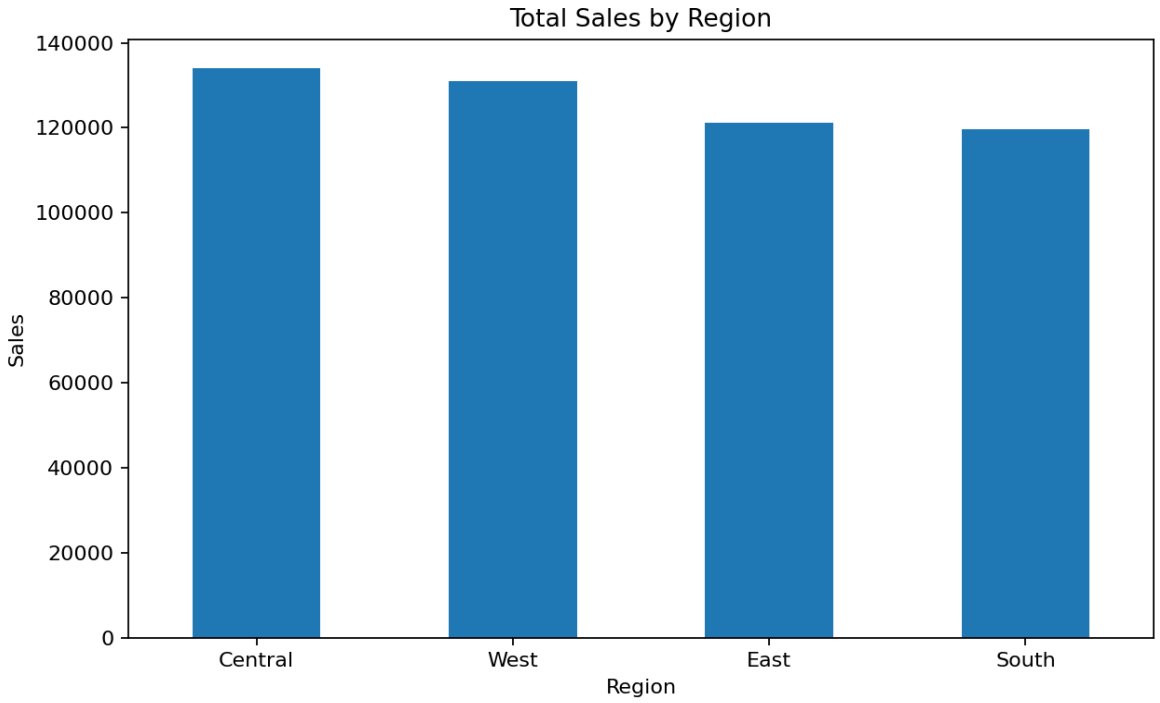
**Dataset:** 500 synthetic practice orders. This is not real company data.

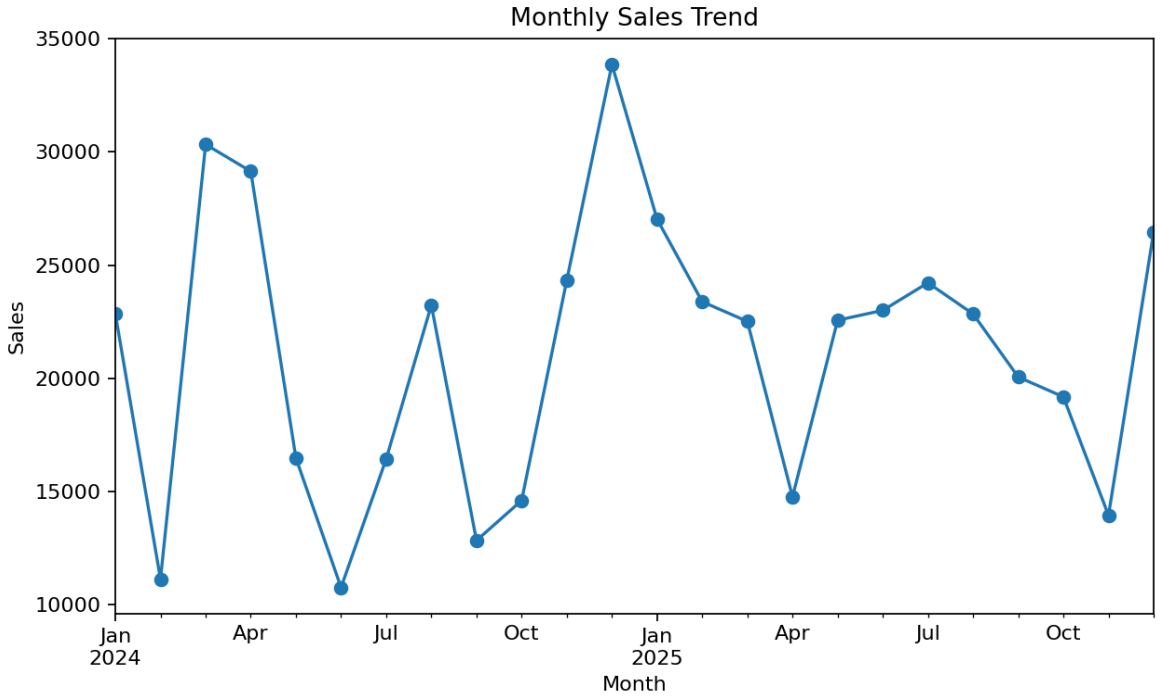
**Total sales:** 505,965.92 **Total profit:** 44,710.77
**Orders:** 500 **Average sales/order:** 1,011.93

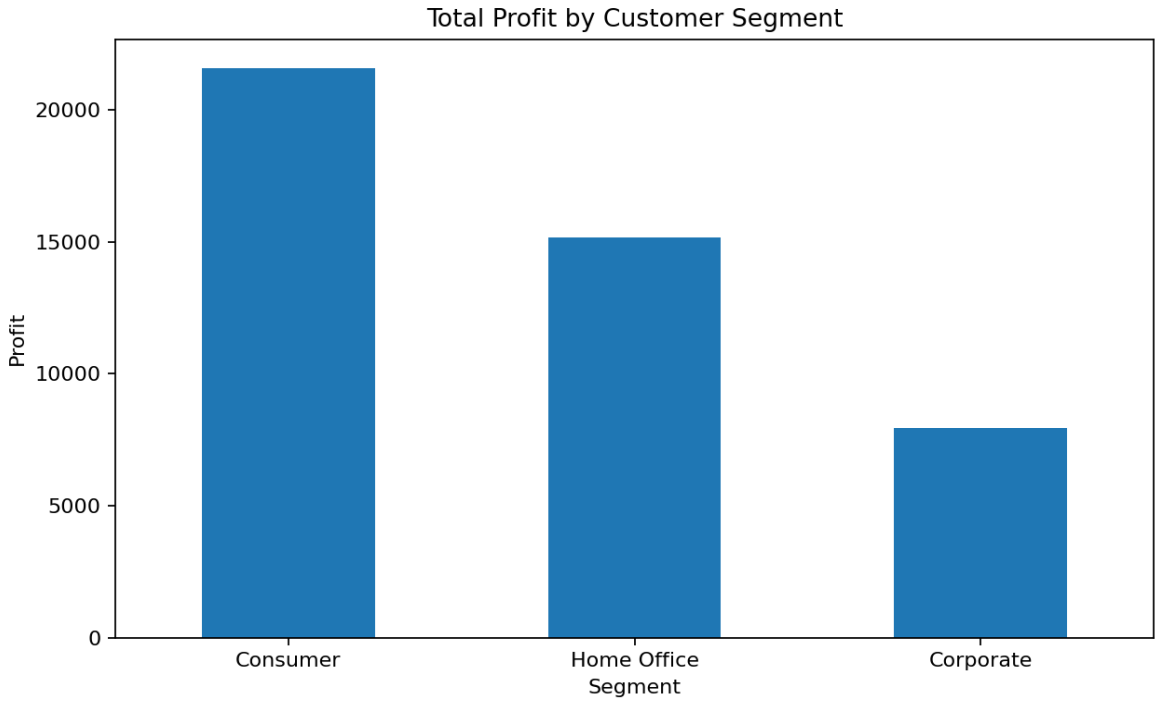
## Key findings

• Technology is the highest-sales category.
• Central is the highest-sales region.
• Consumer is the most profitable customer segment in this sample.
• 154 orders have negative profit and may need review.




Total Sales by Region
140000
120000
100000
80000
60000
40000
20000
0
Sales
Central
West
East
South
Region


Monthly Sales Trend
35000
30000
25000
20000
15000
10000
Sales
Jan
2024
Apr
Jul
Oct
Jan
2025
Apr
Jul
Oct
Month




## Recommendations

Investigate the drivers behind strong category and regional performance; review low-performing months; and inspect loss-making orders. Compare profit as well as sales when evaluating performance.

## Method and limitation

The analysis uses pandas for grouping and aggregation and Matplotlib for bar and line charts. The data is randomly generated for learning; insights are illustrative only and must not be treated as actual business findings.